



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CO 3 – ASSIGNMENT 2

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Course Code & CO Reference: CSA0920 – Multithreading

Assignment Set: CO 3 – Assignment 2

Question: Thread Naming and Identification

Submission Date:

GitHub Repository Link: <https://github.com/192424318pradeep-1980/CSA0920-Java>

Question:

Thread Naming and Identification

Write a Java program that creates multiple threads and explicitly assigns each one a custom name using setName() (or the Thread constructor). Inside each thread's run() method, retrieve and print its own name using getName() and its unique thread ID (using getId() or threadId() in newer Java versions), so each thread's output can be clearly identified.

Concepts: getName(), setName(), thread identification.

PROBLEM UNDERSTANDING

This program creates multiple threads and assigns each thread a custom name using `setName()`. Inside the `run()` method, each thread prints its own name using `getName()` and its unique thread ID using `getId()`. The purpose is to understand how to identify and manage multiple threads in Java.

APPROACH / DESIGN NOTES:

- Create a class that extends `Thread`.
- Assign a custom name to each thread using `setName()`.
- Override the `run()` method.
- Inside `run()`, print the thread name and thread ID.
- Start all the threads using `start()`.

Alternative Considered:

Threads can also be created by implementing the `Runnable` interface, but extending the `Thread` class is simpler for this program.

SOURCE CODE:

```
class MyThread extends Thread {  
    public void run() {  
        System.out.println("Thread Name : " + getName());  
        System.out.println("Thread ID   : " + getId());  
        System.out.println();  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        MyThread t1 = new MyThread();  
        MyThread t2 = new MyThread();  
        MyThread t3 = new MyThread();  
        t1.setName("Thread-A");  
        t2.setName("Thread-B");  
        t3.setName("Thread-C");  
        t1.start();  
        t2.start();  
        t3.start();  
    }  
}
```

}

Expected Output:

Thread Name : Thread-C

Thread ID : 17

Thread Name : Thread-B

Thread ID : 16

Thread Name : Thread-A

Thread ID : 15

TEST CASES:

S.No	Input / Operation	Expected Output	Actual Output
1	Create Thread-A	Displays Thread A and its Thread ID	Success
2	Create Thread-B	Displays Thread-B and its Thread ID	Success
3	Create Thread-C	Displays Thread-C and its Thread ID	Success

SCREENSHOTS OF EXECUTION:

Test Case 1: Thread-A

```
Thread Name : Thread-A  
Thread ID   : 15
```

Test Case 2: Thread-B

```
Thread Name : Thread-B  
Thread ID   : 16
```

Test Case 3: Thread-C

```
Thread Name : Thread-C  
Thread ID   : 17
```

REFLECTION:

I learned how to create multiple threads in Java, assign custom names using `setName()`, and identify each thread using `getName()` and `getId()`. This program helped me understand basic thread management and identification.

DECLARATION:

I declare that the work presented in this assignment is original and completed by me. Any external resources or AI assistance used during the preparation of this assignment have been properly acknowledged.